

A STABILIZER CODE ADMITTING NO LOGICAL Z THAT MEETS THE HYPOTHESIS OF ZHENG–LIU PROPOSITION 3

ABSTRACT. Zheng and Liu map a magic-state distillation protocol to a dynamical system on the Bloch coordinates (x, y, z) . Their Proposition 3 shows that the plane $z = 0$ is a fixed plane of the map provided that $s_j \bar{Z}_i$ contains at least one tensor factor Z for every logical operator $\bar{Z}_i$ and every element s_j of the stabilizer group. Immediately after that proposition they conjecture, in a hedged sentence of unnumbered running text, that the hypothesis can always be met, for any stabilizer code, by properly choosing the logical Z operators. For the $[[3, 1, 1]]$ code $S = \langle \mathbf{XZI}, \mathbf{ZXX} \rangle$ used by the same paper, each of the four cosets of S in its normalizer contains a Pauli string with no tensor factor Z , so no choice of logical Z meets the condition. Proposition 3 itself is correct and is untouched; the map is never computed here, and nothing here shows that this code lacks an invariant coordinate plane. What fails is only the assertion that the hypothesis of Proposition 3 can always be arranged.

1. THE CLAIM

Let $\mathcal{P}_n$ be the group of n -qubit Pauli operators. For a stabilizer code with abelian stabilizer group $S \leq \mathcal{P}_n$ of rank $n - k$ [2], write $\mathcal{N}(S)$ for the normalizer of S in $\mathcal{P}_n$ and call the nontrivial cosets of S in $\mathcal{N}(S)$ the *logical classes* of the code; there are 4^k cosets in all, since $|\mathcal{N}(S)| = 2^{n+k}$ and $|S| = 2^{n-k}$ modulo phases.

Zheng and Liu [5] send a stabilizer-code magic-state distillation protocol to a polynomial map of the Bloch vector of the input state — a viewpoint going back to Rall [4], on top of the structural theory of stabilizer-code distillation protocols of Campbell and Browne [1]. Their Proposition 3 reads, verbatim:

For a stabilizer code Q , $z = 0$ is a fixed plane in the mapped dynamical system if $s_j \bar{Z}_i$ explicitly contains Pauli Z operators for every logical operator $\bar{Z}_i$ ($i = 1, \dots, k$) and every stabilizer element s_j ($j = 1, \dots, 2^{\bar{n}}$).

Here $\bar{n} = n - k$, so the index j runs over the whole group S and not over a generating set. Immediately after the two-line proof of that proposition, the following sentence appears:

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This condition is satisfied by most known protocols that we have examined and *we conjecture it to be generally true for any stabilizer codes by properly choosing the logical Z operators without modifying the stabilizer groups.*

It is the universal form of this sentence that is at issue below. Three conventions of [5] fix its content, and we record them because the counterexample is sensitive to all three.

(i) “Contains Pauli Z ” is about tensor-factor type, and a Y factor does not count. The proof of Proposition 1 of [5] writes a stabilizer element as $s_j = (-1)^{\alpha_j} s_j^1 \otimes \cdots \otimes s_j^n$ with $s_j^\ell \in \{I, X, Y, Z\}$, and the paper then counts the weights w^X, w^Y, w^Z of $\overline{Z}_i s_j$ separately, Y contributing to the Bloch monomial y and not to z . The factorisation $T_i^z = z \cdot \tilde{T}_i^z$ that the proof of Proposition 3 needs therefore requires a genuine factor Z .

(ii) Phases are irrelevant, for the same reason: only the multiset of tensor-factor letters is read. We accordingly work in $\mathcal{P}_n$ modulo phases throughout, identifying a Pauli operator with its string over $\{I, X, Y, Z\}$; the product of two strings is the factorwise product, e.g. $XZI \cdot ZXX = YXX$.

(iii) “Without modifying the stabilizer groups” fixes S . A legitimate choice of logical Z operators is then k elements $\overline{Z}_1, \dots, \overline{Z}_k$ of $\mathcal{N}(S)$ that pairwise commute and are independent modulo S .

Definition 1. A Pauli string is Z -free if none of its tensor factors is Z , i.e. if it is a string over $\{I, X, Y\}$. A coset $QS \subseteq \mathcal{N}(S)$ is *dirty* if it contains a Z -free string, and *clean* otherwise.

Lemma 2 (the coset reduction). *Fix $Q \in \mathcal{N}(S)$. Then Q satisfies the hypothesis of Proposition 3 — that $s_j Q$ contains a factor Z for every $s_j \in S$ — if and only if the coset QS is clean. In particular the hypothesis depends only on the class of Q in $\mathcal{N}(S)/S$, and multiplying Q by an element of S buys nothing.*

Proof. As s_j runs over S the product $s_j Q$ runs over the coset QS exactly once, because S is a group. So “every $s_j Q$ contains a factor Z ” says precisely that no member of QS is Z -free. The identity lies in S , so Q itself is one of the strings tested. $\square$

By Lemma 2, “properly choosing the logical Z operators” is a choice among the $4^k - 1$ nontrivial logical classes: the conjecture asserts that every stabilizer code admits k pairwise commuting, independent-modulo- S elements of $\mathcal{N}(S)$ whose classes are all clean. We call such a k -tuple a *clean frame*. Note that the trivial class S is always dirty, since the identity string is Z -free; and for a code with no clean class at all, not merely does no valid frame exist but no single $\overline{Z}_i$ does.

2. A CODE OF THE SOURCE PAPER FOR WHICH NO CHOICE WORKS

Theorem 3. *Let $S = \langle XZI, ZXX \rangle$, the stabilizer group of the $[[3, 1, 1]]$ code worked out in [5]. Then $S = \{III, XZI, ZXX, YYX\}$, $|\mathcal{N}(S)| = 16$, and $\mathcal{N}(S)/S$ has four classes, each of them dirty:*

$$\begin{aligned} \text{class of } III: & \quad III \cdot III = III, & \text{class of } IIX: & \quad IIX \cdot III = IIX, \\ \text{class of } IZZ: & \quad IZZ \cdot YYX = YXY, & \text{class of } IZY: & \quad IZY \cdot XZI = XIY. \end{aligned}$$

Hence no $Q \in \mathcal{N}(S)$ satisfies the hypothesis of Proposition 3 for this code, so the universal claim of the conjectural sentence quoted in Section 1 fails.

Proof. The two generators commute — their symplectic product is $(1, 0, 0) \cdot (1, 0, 0) + (0, 1, 0) \cdot (0, 1, 1) = 1 + 1 = 0$ over $\mathbb{F}_2$ — and are independent, so S is an abelian group of order 4 with $XZI \cdot ZXX = YYX$, and the code is a $[[3, 1]]$ code, $k = 1$. The published logical pair of [5] is $\bar{Z} = IIX$, $\bar{X} = IZZ$; both lie in $\mathcal{N}(S)$, neither lies in S , and they anticommute, so they are a genuine conjugate pair and the third nontrivial class is that of $\bar{Y} = \bar{Z} \bar{X} = IZY$. For $k = 1$ these three classes together with S exhaust $\mathcal{N}(S)/S$, since $4^k = 4$.

Each displayed line is an identity in $\mathcal{P}_3$ modulo phase whose multiplier is an element of S , and each right-hand side is a string over $\{I, X, Y\}$, i.e. Z -free. So each of the four cosets contains a Z -free string and is dirty. By Lemma 2 no $Q \in \mathcal{N}(S)$ has $s_j Q$ containing a factor Z for all $s_j \in S$. A fortiori no clean frame exists. The stabilizer group has not been modified. $\square$

3. SCOPE

Proposition 3 is a sufficient condition, stated with “if” and never with “iff”, and it is correct; nothing above touches it. What Theorem 3 contradicts is only the separate, unnumbered and hedged sentence asserting that the hypothesis of Proposition 3 can always be arranged, and in that sentence only the universal quantifier over stabilizer codes. In particular nothing above says that $z = 0$ fails to be a fixed plane of the map for this code: we do not compute the map at all. Nor is the coordinate-plane route to a fixed plane closed for it, since the analogue of the condition in the y direction — every member of the coset carries a factor Y — is satisfied by exactly one of its four classes. The other codes and protocols of [5] are not examined here, and nothing here contradicts any numerical result of [5].

The witness code is not ours: [5] takes it from Howard and Dawkins [3] and reproduces it verbatim, and what is new here is the statement about it rather than the object.

4. REPRODUCTION AND VERIFICATION

Theorem 3 is a short computation in $\mathcal{P}_3$ modulo phase: encode a length- n string as the pair $(a, b) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$ with $a_i = 1$ iff factor i is X or Y and $b_i = 1$ iff it is Z or Y ; the product is $(a \oplus a', b \oplus b')$, two strings commute iff $\langle a, b' \rangle + \langle b, a' \rangle = 0$ over $\mathbb{F}_2$, and a string contains a factor Z iff $\bar{a} \wedge b \neq 0$.

The files `verify.py` and `verify.output.txt` accompanying this note are a program and a recorded run of it. From literals only, with no external data, in exact integer arithmetic and with no randomness, it checks for the code of Theorem 3 that S is abelian with the printed element list, that $|\mathcal{N}(S)| = 2^{n+k}$ and $|\mathcal{N}(S)/S| = 4^k$, that each displayed product closes as Pauli arithmetic with a multiplier genuinely in S and a Z -free right-hand side, and that the four representatives lie in pairwise distinct cosets and so exhaust the classes; independently of those certificates it then sweeps all of $\mathcal{N}(S)$ and counts the classes on which the condition holds, in each of the three coordinate directions. Those are the checks that bear on this paper. The recorded run also carries checks on other codes and statements, and its labels and commentary refer to sections, lemmas, propositions and tables of a longer draft rather than to anything printed here; none of that is claimed in this paper. Nothing in the program computes a dynamical map: what it decides is the satisfiability of the hypothesis of Proposition 3, not whether $z = 0$ is a fixed plane.

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